

## Chapter three

# Investment and Saving

### OUTLINES

- Meaning of capital and investment
- Determinants of Investment in Less Developed Countries
- The Rationales and Decision Criteria for Investment
- Investment Demand and Saving Curve
- Theories of Investment
  - Keynesian marginal efficiency of capital (MEC)
  - Accelerator theory of investment
  - Internal fund theory of investment
  - Tobin q – theory of investment
  - Neo-classical theory of investment

# Meaning of capital and investment

- **Capital** is also called as all the man-made goods that are used in the further production of wealth. Reflecting on this we can also call **capital as a man-made resource of production.**
- **Investment** is a crucial component of any economy as the growth of the economy depends on the level of investment.
- **Investment** can be defined as the process of putting ones resource (money) in a given system with expectation of some benefits(more income, some products for sale or for consumption and satisfaction).

## Some examples of investment

- Establishing a production plant
- Opening a new business
- Expanding an existing business
- Constructing public infrastructures such as road schools and hospitals
- Building business centers or residential houses and so on.

- Investment is generally carried out for a variety of reasons. Hence the determinant of investment **is not** a single factor. Some of the major factors that affect investment decision are
- Market demand (required for investment in production)
- Availability of enough financial resource
- Political factors (political stability)
- Level of uncertainty (level of risk)
- Availability and efficiency of banking system
- Government economic or investment policy
- Interest rate (cost of borrowing)
- The size of liquid assets at dispose of the investor

# The Rationales and Decision Criteria for Investment

- Why did individuals and government make investment?
- There are two major reasons
  - ✓ Profit motive
  - ✓ Non profit motive (the welfare reason or humanitarian issues)

## Decision criteria's for investment

### 1. Present value criteria

$$NPV = \sum_{t=0}^N PVB - \sum_{t=0}^N PVC$$

$$NPV = \sum_{t=0}^N \frac{B_t}{(1+r)^t} - \sum_{t=0}^N \frac{C_t}{(1+r)^t}$$

# Decsion rule

## SINGLE PROJECT Decision criteria:

- Accept investment: if the NPV is positive
- What positive NPV means is that the Return from the investment is more than the Risk inherent in the investment or
- A positive NPV indicates that the investment will lead to an increase in the wealth to the investor/project owner
- Reject investment: if the NPV is negative (that is the risk is more than the return)

## TWO OR MORE COMPETING PROJECTS Decision criteria:

- Accept investment: with the **highest NPV** and such NPV is **positive** otherwise reject the investment .

## 2. Marginal efficiency criteria (Internal Rate of Return)

- IRR - is the discount rate at which an investment has a **zero net present value**.
- Allows the risk associated with an investment project to be assessed
- The **internal rate of return** **equate** to **the interest rate**, expressed as a percentage.
- Therefore, if the internal rate of return for the project is less than the **current bank interest rate** it would be **more profitable** to put the money in the bank than execute the investment project

## Cont'd

Decision rule – if  $IRR > \text{Cost of capital}$  accept the project  
- if  $IRR < \text{Cost of capital}$  Reject a project

The formula for calculating the Internal Rate of Returns (IRR) is expressed as:

$$IRR = L + C (NPV1/E)$$

Where:

L = Lower Discount rate

C = Difference between the two discount rate

NPV1= net Present value of the lower discount rate

NPV2= Net Present value at higher discount rate

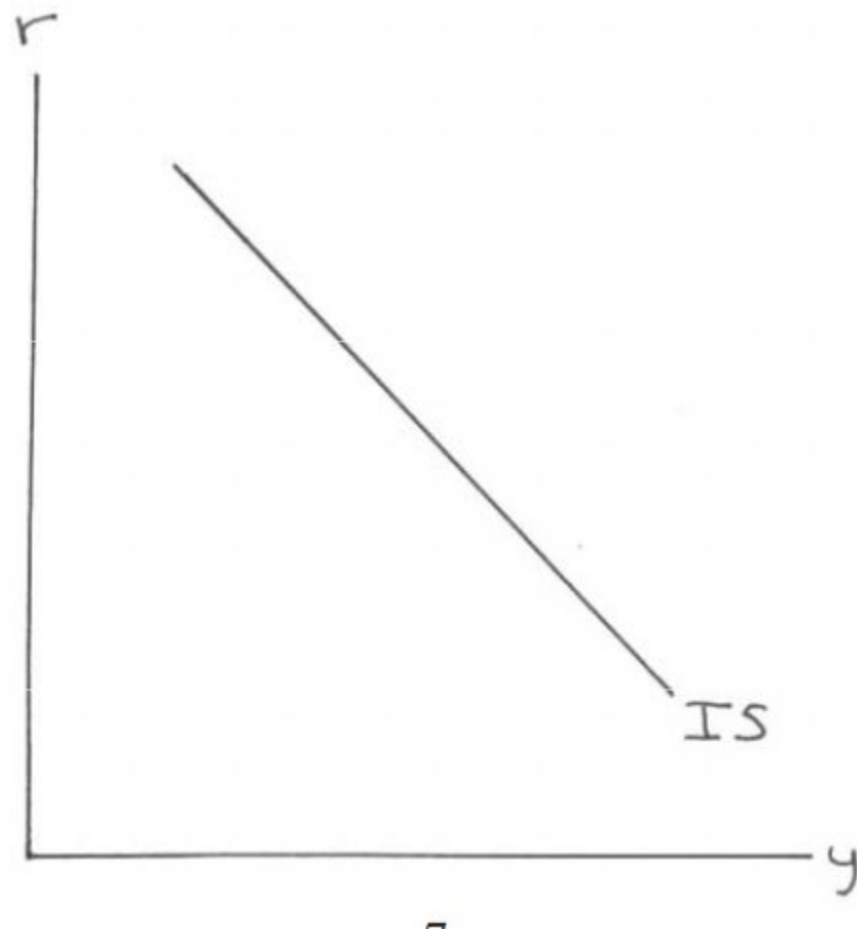
$E = |NPV1| + |NPV2|$

# Investment Demand and Saving Curve

- **Investment Demand**

The interest rate is the cost of capital to the firm. We model real investment demand  $i(r)$  as a decreasing function of the real interest rate  $r$  (figure 1). As the cost of capital rises, the investment demand falls.

Figure 2: IS Curve



- **Aggregate Demand Equals National Product**  
That aggregate demand equals national product means that consumption demand  $c(y)$  plus investment demand  $i(r)$  plus government expenditure on goods and services  $g$  equals national product  $y$ :

$$c(y) + i(r) + g = y$$

- Desired Investment Equals Desired Saving  
The name “IS curve” derives from the property that it represents that desired investment equals desired saving.

Let  $t$  denote taxes. Rearranging (1) gives

$$i(r) = [y - t - c(y)] + (t - g) .$$

The left-hand side is desired investment.

- The right-hand side is desired saving:  $y - t - c(y)$  is household saving (disposable income  $y - t$  less consumption demand), and the government surplus  $t - g$  is government saving.

# IS Curve

Think of the national income and product  $y$  and the interest rate  $r$  as defining the state of the economy. Given these two variables, one can determine the aggregate demand.

In figure 2, the IS curve shows the combinations of  $y$  and  $r$  such that aggregate demand equals national product.

## **Downward-Sloping IS Curve**

The IS curve is downward sloping. When the interest rate falls, investment demand increases, and this increase causes a multiplier effect on consumption, so national income and product rises.

## **Flat or Steep?**

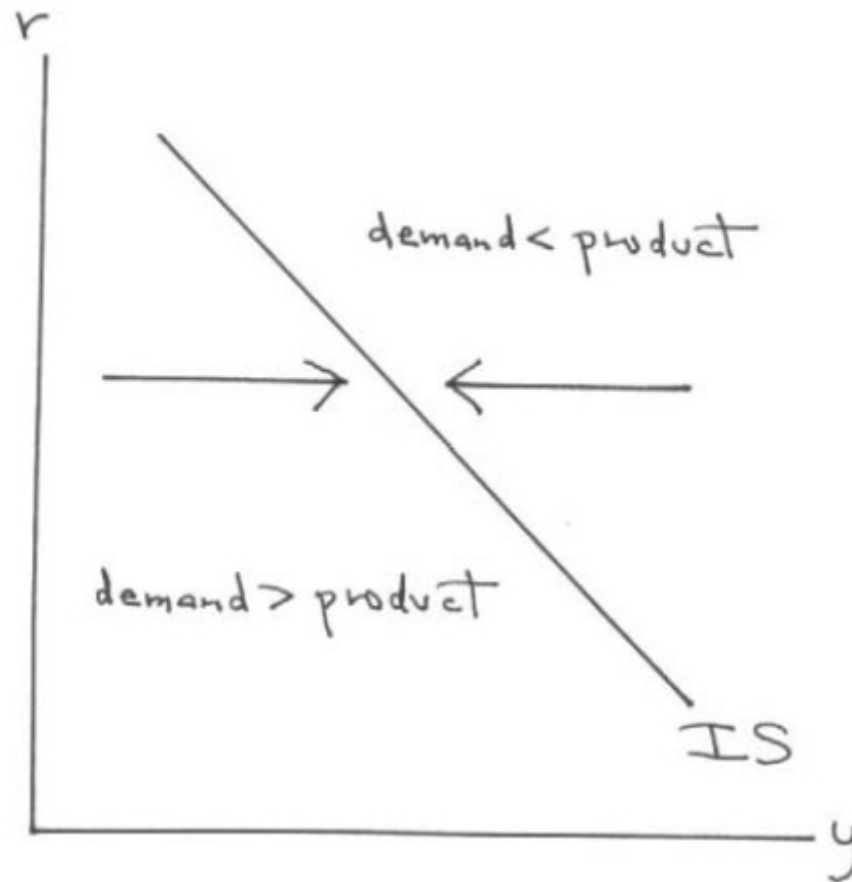
If investment demand is highly sensitive to the interest rate, then a reduction in the interest rate causes a big increase in national income and product. Hence the IS curve is flat.

If the marginal propensity to consume is high, then a given change in investment demand causes a big increase in national income and product. Hence the IS curve is flat.

# Adjustment to the IS Curve

- National product adjusts to put the economy on the IS curve in the short run (figure 3).
- To the left of the IS curve, aggregate demand exceeds the product, so firms expand production to meet demand.
- To the right, aggregate demand is less than the national product. Firms reduce production, since they will not produce what they cannot sell.

Figure 3: Adjustment to the IS Curve



# Theories of Investment

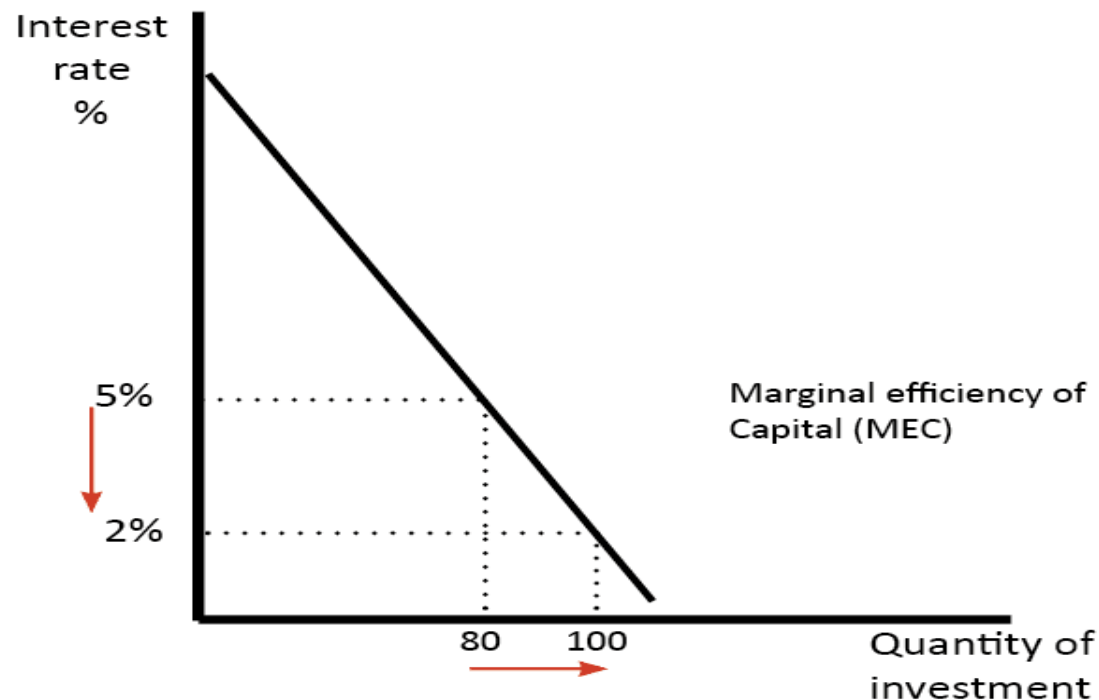
- **Keynesian marginal efficiency of capital (MEC)**

- The marginal efficiency of capital displays the expected rate of return on investment, at a particular given time. The marginal efficiency of capital is compared to the rate of interest.
- Keynes described the marginal efficiency of capital as:
  - “The marginal efficiency of capital is equal to that rate of discount which would make the present value of the series of annuities given by the returns expected from the capital asset during its life just equal to its supply price.”

- This theory suggests investment will be influenced by:
  - ✓ The marginal efficiency of capital
  - ✓ The interest rates
- Generally, a lower interest rate makes investment relatively more attractive.
- If interest rates, were 3%, then firms would need an expected rate of return of at least 3% from their investment to justify the investment.
- If the marginal efficiency of capital was lower than the interest rate, the firm would be better off not investing, but saving the money.

## ❑ Why are interest rates important for determining the marginal efficiency of capital?

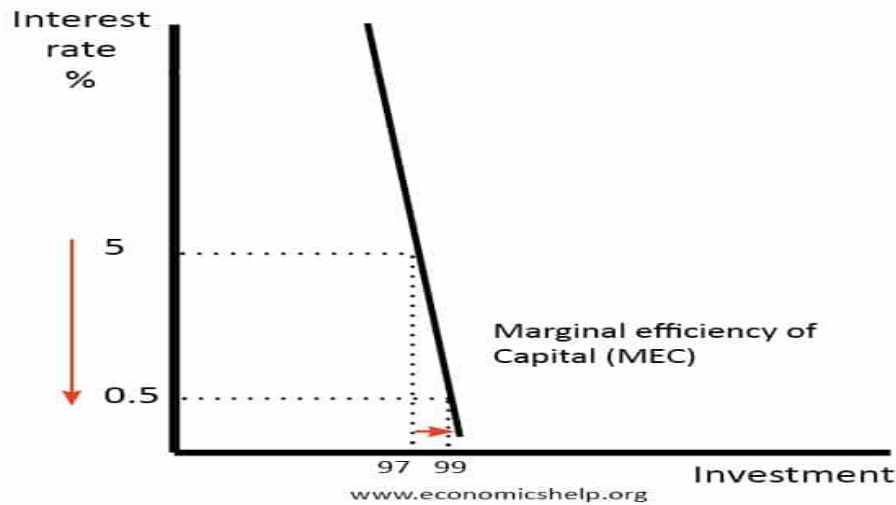
- To finance investment, firms will either borrow or reduce savings. If interest rates are lower, it's cheaper to borrow, or their savings give a lower return making investment relatively more attractive.



- A cut in interest rates from 5% to 2% will increase investment from 80 to 100.
- The alternative to investing is saving money in a bank; this is the opportunity cost of investment.
- If the rate of interest is 5%, then only projects with a rate of return of greater than 5% will be profitable.

### **How responsive is investment to interest rates?**

- In Keynesian investment theory, interest rates are one important factor. However, in a [liquidity trap](#), investment may be unresponsive to lower interest rates. In some circumstances, demand for investment is very interest inelastic.



- In a liquidity trap, business confidence may be very low. Therefore, despite low-interest rates, firms don't want to invest because they have low expectations of future profits.
- **Liquidity trap** is when monetary policy becomes ineffective because, despite zero/very low-interest rates, people want to hold cash rather than spend or buy illiquid assets.)

## **Factors that can affect investment schedule**

At the same rate of interest rate – more investment projects are demanded. This could reflect an improvement in economic circumstances, which encourage firms to invest.

**1. The cost of capital.** If capital is cheaper, then investment becomes more attractive. For example, the development of steel rails made railways cheaper and encouraged more investment.

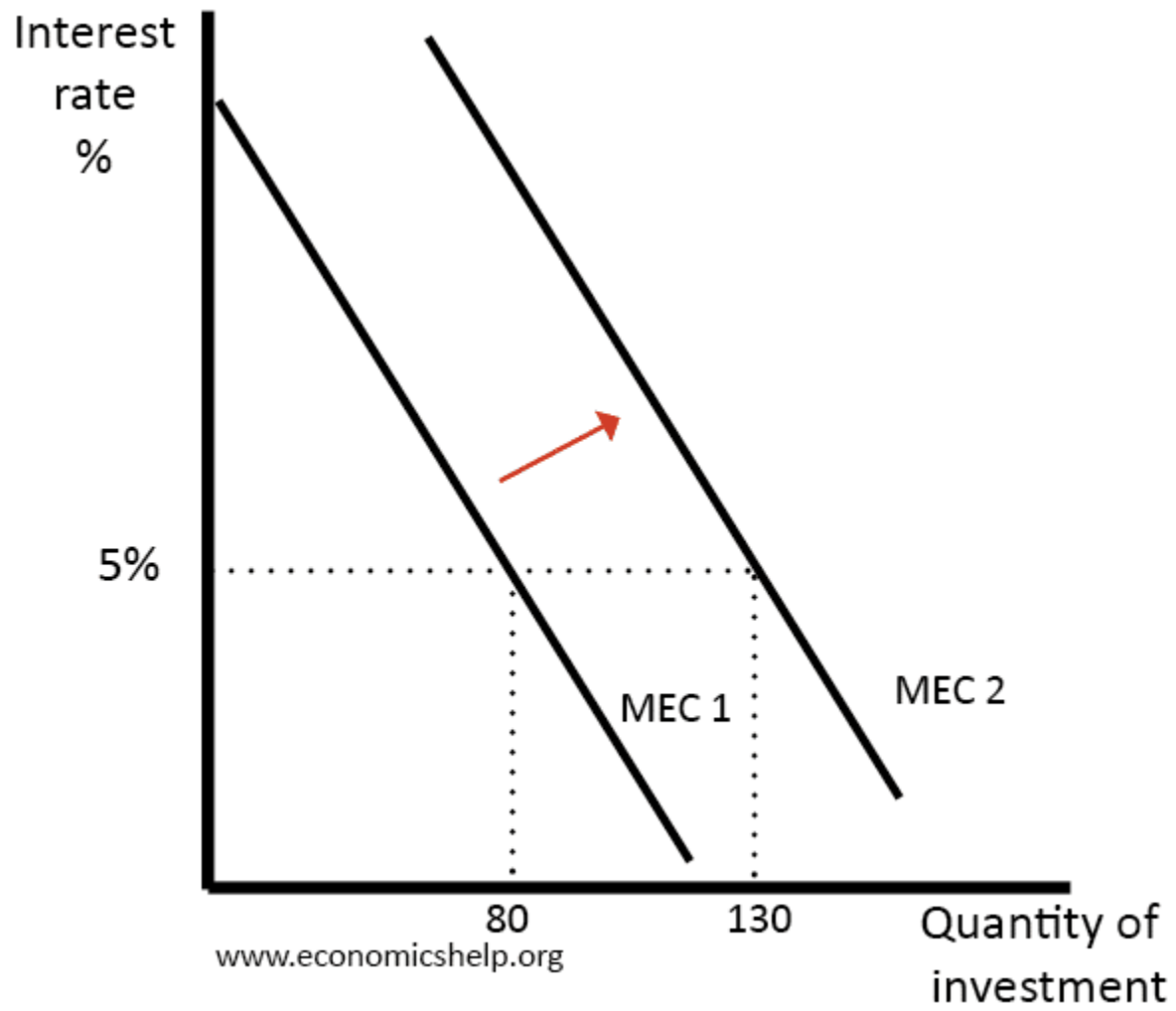
**2. Technological change.** If there is an improvement in technology, it can make investment more worthwhile.

**3. Expectations and business confidence.** If people are optimistic about the future, they will be willing to invest because they expect higher profits. In a *recession*, people may become very pessimistic, so even lower interest rates don't encourage investment.

**4. Supply of finance.** If banks are more willing to lend money investment will be easier.

**5. Demand for goods.** Higher demand will increase the profitability of capital investment.

**6. The rate of Taxes.** Higher taxes will discourage investment. Sometimes, governments offer tax breaks to encourage investment.



# **Accelerator theory of investment**

- The Keynesian concept of multiplier states that as the investment increases, income increases by a multiple amount.
- The acceleration principle describes the effect quite opposite to that of multiplier.
- According to this, when income or consumption increases, investment will increase by a multiple amount.
- When income and therefore consumption of the people increases, the greater amount of the commodities will have to be produced.

- This will require more capital to produce them if the already given stock of capital is fully used.
- Since in this case, investment is **induced** by changes in income or consumption, this is known as **induced investment**.
- The accelerator is the numerical value of the relation between the increase in investment resulting from an increase in income.
- The net induced investment will be **positive** if national income increases and
- induced investment may fall to **zero** if the national income or output remains **constant**.
- To produce a given amount of output, it requires a certain amount of capital. If  $Y_t$  output is required to be produced and  $v$  is capital-output ratio, the required amount of capital to produce  $Y_t$  output will be given by the following equation:

$$K_t = vY_t \dots(i)$$

where

- $K_t$  stands for the stock of capital,
- $Y_t$  for the level of output or income, and
- $v$  for capital-output ratio.

- This capital-output ratio  $v$  is equal to  $K/Y$  and in the theory of accelerator this capital-output ratio is assumed to be **constant**.
- Therefore, under the assumption of constant capital-output ratio, changes in output are made possible by changes in the stock of capital.
- Thus, when income is  $Y_t$  then required stock of capital  $K_t = vY_t$ .
- When output or income is equal to  $Y_{t-1}$ , then required stock of capital will be

$$K_{t-1} = vY_{t-1}.$$

- It is clear from above that when income increases from  $Y_{t-1}$  in period  $t - 1$  to  $Y_t$  in period,  $t$ , then the stock of capital will increase from  $K_{t-1}$  to  $K_t$ . As seen above,  $K_{t-1}$  is equal to  $vY_{t-1}$  and  $K_t$  is equal to  $vY_t$ .
- **Hence, the increase in the stock of capital in period  $t$  is given by the following equation:**

$$K_t - K_{t-1} = vY_t - vY_{t-1}$$

$$K_t - K_{t-1} = v (Y_t - Y_{t-1}) \dots (ii)$$

- Since increase in the stock of capital in a year ( $K_t - K_{t-1}$ ) represents **investment** in that year, the above equation (ii) can be written as below:

$$I = V (y_t - y_{t-1}) \dots (iii)$$

- By definition net investment is equal to the gross investment minus capital consumption allowance or depreciation (D). This can be incorporated in our equation as follows
- $I-D = v(Y_t - Y_{t-1}) = v\Delta Y$
- If aggregate demand is constant then net investment is zero. because net investment is given as follows

$$I-D = v(Y_t - Y_{t-1}) = v\Delta Y \text{ when } \Delta Y=0$$

- $v(0)$
- $I-D=0$  net investment equal to zero.

# Internal fund theory of investment

- The central view of this theory is that investment depends on the *level of profit*. The managers have preference for financing investment internally.
- Firms can obtain finance from two sources internal to the firm and external to the firm .
- If the firm opts for external borrowing then it needs a series of fixed payment to return borrowings.
- During recession, when the economy doesn't perform well the firm may also not perform well and firms may not fulfill the commitment to pay.
- Similarly selling of stock has a chance of losing control. For these reasons the proponent of internal fund theory argues that firms strongly prefer to finance investment internally and that the increased ability of internal funds through higher profits generates additional investment.

- *Hence this investment theory implies that investment level is not determined by the level of interest rate or cost of borrowing, regulating the level of interest rate does not help achieve the required level of investment, rather the level of profit that investors make determines the level of investment.*
- Policy makers of this theory are interested in increasing investment. Internal fund theory holds the view that profits should be increased. *Policy makers can do this by reducing corporate taxes and reduction of income tax.*

# Tobin q – theory of investment

- The firm needs money for investment. This money can be raised either by borrowing or by selling shares, equity, etc.
- When the firm sells the share, the buyer buys the share to earn a capital gain from the increase in the market value of the shares.
- The purchaser of share, therefore, purchases shares when he expects a high capital gain. This is because: The share price is high. The firm by selling only few shares can raise a lot of money.
- Thus when stock markets **are high**, firms are willing to sell equity to finance investment than when the stock market is low.

- James Tobin was the first person to explain this relation between the stock market and investment and that is why it is also referred as “Tobin’s q” theory.
- $q = \text{market value of the firm} / \text{Replacement cost of capital}$

Or

- $q = \text{Value the stock market places on the firm's asset} / \text{Cost of producing those assets}$

(i) If  $q$  ratio is high  $\rightarrow$  it means the price of share is high.

- Firms will invest more.

(ii) If  $q > 1 \rightarrow$  Firm will buy physical capital, that is, add to the capital stock.

- for every dollar worth of new machinery, the firm can sell the stock for  $q$  dollar and earn a profit  $= q - 1$ .
- In other words, when  $q > 1$ , firms find it profitable to acquire additional capital because value of capital exceeds the cost of acquiring it.
- Thus, when  $q > 1 \rightarrow$  Investment will increase.

*The q theory explains the relationship between Investment and the Capital Stock*

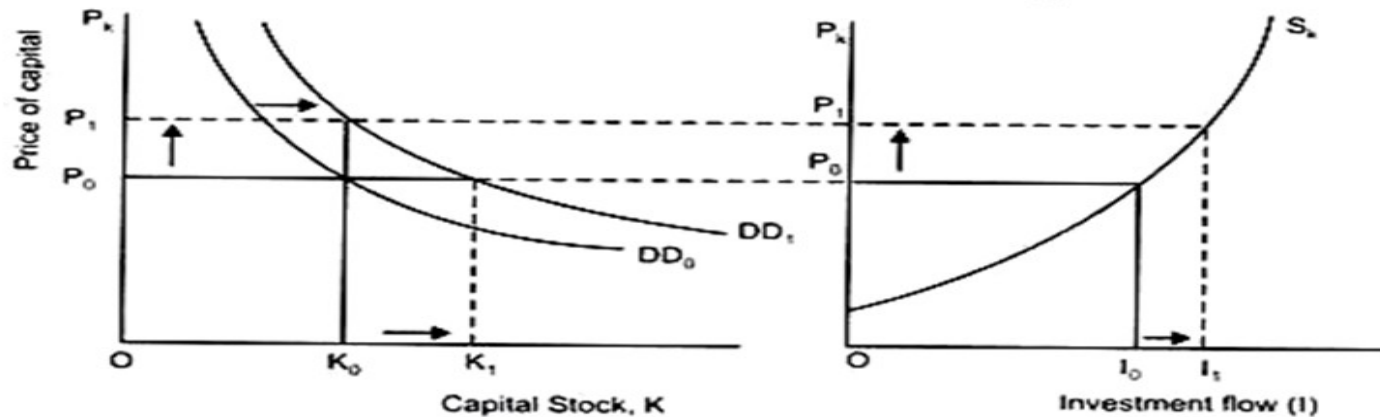


Figure 20.4

(a) Initial capital stock  $\rightarrow K_0$  (Fig. 20.4a)

*Assume demand for desired capital increases from  $K_0$  to  $K_1$ .*

Demand curve of capital shifts to the right from  $DD_0$  to  $DD_1$ .

- Due to increase in demand for capital ( $P_0 \rightarrow P_1$ )
- Although, desired capital stock is  $K_1$  and Investment is  $I_1$ , but, actual capital stock is  $K_0$  & Investment  $\rightarrow I_0$  (Fig. 20.4b)

- Therefore, in the short run there exists a gap between the Actual Capital stock ( $K_0$ ) and the desired capital (that is  $K_1$ ).

### **Reason:**

- Factors of production and other inputs required to produce additional capital are in short supply. Therefore, capital stock cannot be increased immediately in short run.

## **In the long run:**

- The gap between the capital stock ( $K_0$ ) and Investment flow ( $I_1$ ) is filled in long run such that:
- Desired capital stock ( $K_1$ ) = Actual capital stock ( $K_1$ )

## **Neo-classical theory of investment**

- According to neo classical theory of investment, investment is based on benefit and cost of the investment activity to a firm or firms.
- This theory assumes that firms borrow capital at a rate (R) from the owner of the capital and sell its product at a price P.
- Then we can say that the cost of capital is equal to  $(R/P)$ . In terms of benefit, we can get the productivity of the capital, which goes down with more investment at margin.
- Real cost  $= R/P$  this cost has the shape of normal curve function which is increasing or outward sloping.

- Real benefit is measured in terms of marginal productivity of capital(MPk). The curve measuring this benefit is downward sloping since marginal product of a factor of production including capital declines as the level of employment of the factors increases.
- The level of investment has to keep on increasing as long as the benefit of doing it is greater than the cost of doing so or the cost of investment activity. This implies that the investment level should increase up to the level marginal product of the investment good or capital equals to user cost. This is represented by the point of intersection between the cost (R/P) curve and the benefit (MPk)curve.

$$R/P = MP_k$$

$$R = P \cdot MP_k$$

## □ Assignment

- Criticize accelerator theory of investment?

# Chapter four

- **The Money Market**
  - Types of Money
  - Money Supply
    - 4.2.1 Components of money supply
    - 4.2.2. Creation of money and banking system
- 4.2.3. A simple Model of Money Supply
- 4.2.4. Money supply, Near Money and Seigniorage
- 4.2.5. Income Velocity of Money and Quantity
- 4.3 Theory of Demand for Money
  - 4.3.1. Major functions of money
- 4.3.2. Theories of demand for money
- 4.4. Money Market Equilibrium

## 4.1. Types of Money

- **Money:-** is any agreed up on instrument or material to be used as a medium of exchange of goods and services.
  - **Evaluation of money**
- With the evolution of money, the exchange system was developed or evolved over some major phases. Some of these phases are

### **The stage of barter system**

The initial phase of exchange of goods and services existing before the development of any common medium of exchange.

Any commodity could be used for exchange for another as long as the other party making the exchange is willing to take the commodity.

### **The stage of commodity money**

- where some commodities were used as the common medium of exchange  
Eg. Gold, silver molded salt(amole)

### **The stage of legal/tender money**

- Easily manageable instruments legally declared to function as medium of exchange.

### 4.2.1 Components of money supply

Some of the components of money supply popularly used in many countries are

$M1 = \text{currency with public} + \text{demand deposit with the banking system or deposits in checking account} + \text{checks for current conversion and use (travelers checks)}$

$M2 = M1 + \text{deposit in saving account} + \text{money market mutual funds}$

$M3 = M2 + \text{time deposits with the banking system (long time deposits)} + \text{financial securities such as treasury bills, bonds and so on}$

$M4 = M3 + \text{total post office deposits}$

### 4.2.2. Creation of money and banking system

- Change in the money supply occur due to the action of central bank of a country. They acquire assets of various kinds and issue in payment of liabilities or debts.
- The size of a given amount of money supply increases over time because of the fact that the money can be lent on positive interest rate where borrowers pay back to the bank larger amount than the amount borrowed including the interest payment.
- The central bank also generates or keeps some amount of money from deposits with commercial banks. This part of money supply is known as **reserve requirement**.

- Commercial banks can borrow back from this money.
- In Ethiopia, the central bank of Ethiopia controls the supply of money.
- Any central bank has three major monetary policy tools through which it controls the money supply.
- This monetary policy instruments are
  1. reserve requirement
  2. open market operation
  3. Discount rate

## **Reserve requirement**

- Usually households and owners of financial resources save or deposit their money with commercial banks. The central bank requires commercial banks to retain a certain percentage of their deposits as a reserve in central bank.
- Eg suppose the total deposit made in Ethiopian Commercial bank is 100birr, if the reserve requirement of the central bank is 5 %, then Ethiopian Commercial bank has to deposit 5 birr in central bank of Ethiopia and commercial bank can use 95 birr for giving loan to the customers or borrowers.
- If central bank wants to expand money supply in the country, then it should reduce the reserve requirement.

## **Open market operation**

- It means the purchase and sale of government securities by the central bank to maintain the proper level of money supply in an economy.
- In order to increase money supply, the central bank can buy government bonds from public. By doing so it gives money to the public and effectively increasing money supply.
- On the other hand, by selling government securities to the public central bank can reduce the supply of money in the economy.

## Change in discount rate

- A commercial bank that runs out of reserve or money for its operation such as making loans can borrow either from central bank if reserve is in excess or other banks.
- The cost of borrowing from central bank is known as **discount rate**.
- In other words, it is the interest rate that central bank charges from commercial banks that borrow from it to meet temporary needs for reserves or to use the reserves they need to keep with the central bank.

### 4.2.3.A simple Model of Money Supply

This model involves three exogenous variables; monetary base, reserve and demand deposit.

**1. Monetary Base (B);** this is the total amount of money held by the public as currency( C ) under control of ( or decided for what purpose to use by) the public; and the amount kept by banks as reserves (R) its size and use is determined or controlled by central bank.

This relation is given by the following relation

$$B = C + R \dots\dots\dots (1)$$

**2. Reserve requirement (rr) =  $R/D$** ; is the ratio of reserves to the deposit kept with banks. It is determined by business policies.

**3. Currency-deposit ratio =  $cr = (C/D)$** ; this represents the preference of the public ( consumers and producers) about how much money to hold in the form of currency (C ) and how much to hold in the form of demand deposits (D) which is deposited in banks in checking account from which they withdraw when needed.

the sum of the two represents the money supply in circulation either in the form of currency or deposit in banks given by the following relations.

$$M = C + D \dots\dots\dots (2)$$

- We can obtain more precise equation for money supply take the ratio of eqn 1 and eqn 2.

$$M/B = C+D/C+R$$

- By dividing both numerators and denominators by D, we obtain the following

$$M/B = \frac{C/D + D/D}{C/D + R/D} = \frac{cr+1}{cr+rr}$$

- By multiplying both sides by B again we obtain

$$M = \frac{(cr+1)}{cr+rr} B$$

- Letting  $\frac{cr+1}{cr+rr} = m$ , the simple money supply is given as

$$M = mB = m(C+R) \text{ where } M = \text{money supply and } m = \frac{cr+1}{cr+rr} \text{ called money multiplier}$$

- The meaning of the multiplier is that each birr of monetary base produces m birr of money. When B increases by a unit money supply increases by m factor or unit.
- The lower the reserve deposit ratio(rr), the more loan banks would make and the more money they would create from every birr. This means, a decrease in rr leads to an increase in money supply.
- For instance, if currency deposit ration=0.4 (currency =40% of total deposit) and reserve requirement =10% of deposit then
- $$m = \frac{cr+1}{cr+rr} = \frac{0.4+1}{0.4+0.1} = 2.8$$
- And the simple money supply function is given by  $M=2.8B$

The three major components / determinants of money supply in this simple money supply model ( $B$ ,  $cr$  and  $rr$ ) are again affected by different factors.

- $B$  is affected by government action through  $R$ .
- The value of  $rr$  is also determined by the act of the government.
- $Cr$  is determined by the households behavior.
- This implies that money supply is not determined by government decision alone but also by households' decision.
- For instance, if  $hh$  decides to deposit more money instead of holding, money supply increases even if government takes no action. This is because, this makes the money circulate more and multiply more.

#### 4.2.4. Money supply, Near Money and Seigniorage

- Money supply consists of assets that have acquired liquidity( the ability to purchase goods and services) such as bank cheques. The development of near money causes instability in money demand and finally gives wrong signal about aggregate demand.
- Thus it may mislead policy makers. One solution to such problem is use of a broad definition of money which incorporates these near money as part of the money supply.
- **Seigniorage:-** is the revenue from printing money. When the government or central bank prints money, it increases money supply. Such an increase in money supply usually causes inflation.

## **4.3 Theory of Demand for Money**

### **4.3.1. Major functions of money**

#### Major functions of money

##### 1. transaction function

- For exchange purpose

##### 2. money as a unit of account

- Measuring outputs in terms of money

##### 3. money as store of value

- Converting other assets in to money and kept for future)

### **4.3. Theory of Demand for Money**

#### **A. Classical theory of money demand**

People need or demand money solely for transaction purpose.

#### **B. Keynesian theory of money demand**

People need or demand money for speculative purpose and precautionary demand for money in addition to transaction demand.

#### **C. Portfolio theory of money demand**

The role of money as store of value.

People hold money as part of their portfolio of assets.

Thus, the demand for money depends on the risk and return offered by money and by various other assets.

$M_d/P = L(r_s, r_b, \pi, W)$      Where  $r_s$  is expected return on assets

$r_b$  is expected return on bond

$\pi$  is expected inflation rate

$W$  is wealth

# Money Market Equilibrium

- It is determined by interaction between the level of money supply determined by central bank and money demand which is determined by different factors such as **interest rate** and **income**.
- Total money demand in real terms can be expressed as  $M_d = f(Y) + f(r)$
- $M_s = M_d$